

1 Energy Management

Definitions

- Consumption** Consumption means the withdrawal of raw material and semi-finished products for the production process. They are consumed by the process. Such consumption also includes energy consumption in any form: electric energy, gas, oil, compressed air, water in turbines, etc.
- Since energy actually cannot be consumed, this is indeed more precisely called energy demand.
- The consumption in HYDRA is "measured", in kWh, m³, l or other quantities convertible to joule, according to the relevant medium.
- Note: The colloquial expression "to consume current" is technically not correct, since the current supplied to a device will leave it again (principle of charge conservation). What actually "flows" is electric energy! However, this is not in fact consumed but converted according to the principle of energy conservation. The work performed in this process (the product of voltage, amperage and time in the case of ohmic recipients) is measured by a so-called electric meter and hence represents the consumption of electric energy.
- Electric work** Electric work is the physical term for the transmitted quantity of electric energy and hence synonymous for the energy industry. This means the work specifies the energy demand. The unit of electric work can generally be indicated in joule, "J". In electric energy engineering, the unit "kWh" is used. Conversion: (1 kWh corresponds to 3600 kJ).
- The electric meters connected in HYDRA directly supply the electric work and hence the current "consumption".
- Energy demand** Amount of energy to be put through for work performance - please also refer to "Electric work" or "Consumption". The energy demand is frequently related to a period, e.g. a year. In statistics, the primary energy demand is, for example, indicated in PJ (petajoule). Both electric energy and energy generated by other means is measured by this. In Germany, for example, 14,000.00 PJ of energy were required in 2010 (approx. 3,900 bn kWh).

Energy	Energy is a physical quantity identified by the symbol "E". Energy may be measured in a variety of units, depending on the application. Since energy is related to the work performed for energy conversion, both energy and work may be indicated in joule, kWh, Nm, eV, or even in kg (according to Einstein). The electric energy is calculated from $U \cdot I \cdot t$, i.e. the product of amperage, voltage and time. In addition, energy and/or work is the integral of capacity in mathematical terms. This also explains the connection between capacity meters and consumption meters! According to the principle of energy conservation, energy is not consumed but only converted. The required work corresponds to the amount of converted energy.
Current	In technical terminology, "current" is the designation of its strength. More precisely, it is the physical quantity "amperage" designated by the symbol I and the unit ampere (A). In colloquial language, it is generally used to specify the transmission of electric energy, which tends to result in confusion! Amperage thus describes the amount of electric charge transported in a specific period of time. In the context of HYDRA, current is a measured physical process quantity expressed in ampere, i.e. it is recorded by the mechanisms of the PDV integrated in the EMG module.
Electric meter	The electric meter measures the electric work determined by the connected consumers! This means the energy flow. The unit counted by the electric meter is kWh, i.e. "kilowatt hours". Some devices also provide power values.
Voltage	Electric voltage is a technical quantity and is measured in Volt (V). It describes the potential difference of electric charge between two conductors and is hence practically the power for the electric current, or, more precisely, the work capacity of the electric charge in the electric field. Regarding the specification, a distinction is to be made between constant direct current voltage and alternating voltage, where voltage changes sinusoidally over time. The domestic mains supply in Germany operates with 220 - 230 V (effective value) AC. As from 50 V AC, the insulation capacity of dry skin is overcome so that contact causing ventricular fibrillation may be lethal. For more information, please refer to the relevant literature from physics, electrical engineering and medicine.

Capacity Capacity is a physical quantity which, in simple terms, is calculated as the quotient of work (or expended energy) per unit of time. The unit for capacity is Watt ("W") or multiples of it. Capacity may be indicated for consumers as for energy generators (megawatt for power stations). The instantaneous capacity or power means the energy demand when the time interval moves towards zero. In mathematical terms, this is calculated as differential quotient or 1st derivative. As regards electric capacity, the capacity is calculated as the product of voltage and amperage ($U \cdot I$). This illustrates the relationship between energy and capacity. Since alternating currents are often used in energy engineering, and because this value is a quantity fluctuating across the phases, a mean value is obtained; this is then called effective power, reactive power or apparent power. The effective power is the capacity which can be used by consumers.

In HYDRA, the capacity is measured by connected technical recording devices, in the same way as the amperage or voltage, for example.

As regards non-electrical energy sources, the consumption per unit of time can be measured as capacity, e.g. by means of flow-through scales and/or counters, or by "speed-measuring" sensors.

Since an average capacity over a defined period can be calculated from consumption very easily, it is possible to convert consumption figures into capacities in HYDRA.

Energy efficiency Key figure for measuring the specific energy demand per utility. The key figures are defined individually per company or product, since the utility has to be defined differently in each case. Energy efficiency can, for example, be measured in energy demand per unit produced. In contrast to this, the efficiency is a purely technical quantity and describes the quotient from energy emitted to energy supplied.

Energy management Energy management is the provident, organized and systematic coordination of procurement, conversion, distribution and utilization of energy to cover the requirements in consideration of ecological and economic objectives (pursuant to VDI 4602).

Energy management system According to ISO 50001, an energy management system comprises: The entirety of related or interacting elements for the introduction of an energy policy and strategic energy objectives, as well as of processes and procedures for achieving such strategic objectives.